

HUMBER POLYTECHNIC

Faculty of Applied Sciences & Technology

AIGC-5005: Capstone Project Proposal

Deep Learning-Based Identification of SSB-PriA Interaction Inhibitors Against
Multidrug-Resistant *Klebsiella pneumoniae* Using Graph Neural Networks

Industry Sponsor: Living In Silico Inc.

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Date of Submission: March 22, 2026

1. Title:

Deep Learning-Based Identification of SSB-PriA Interaction Inhibitors Against Multidrug-Resistant *Klebsiella pneumoniae* Using Graph Neural Networks

2. Abstract:

The world's biggest health problem is Antibiotic resistance. *Klebsiella pneumoniae* bacteria are very hard to treat as they are resistant to most antibiotics. This necessitates the development of new antibiotics that act differently to treat *Klebsiella pneumoniae*. Research shows the interaction between two bacterial proteins, SSB (single-stranded DNA binding protein) and PriA helicase. These two are proteins, and they work together to help bacteria to create DNA and spread. Without these two proteins, bacteria cannot survive. This makes the SSB-PriA interaction a strong target for new antibiotics.

The goal of our project is to identify molecules that can block SSB-PriA interaction in *Klebsiella pneumoniae* bacteria. Blocking the interaction can prevent the spreading and development of new bacteria. Instead of testing multiple compounds in the lab, we create model GNNs (Graph Neural Networks) using deep learning to predict which molecules work on the interaction of the SSB-PriA protein.

We target two GNN models: Graph Convolutional Networks (GCN) and Graph Isomorphism Networks (GIN), and compare how well each one performs. We used the AID 1272365 dataset, which contains over 440,000 chemical compounds.

We used RDKit to process and represent molecules as graphs so the models can better understand their structure. We expect our model to become more accurate in identifying the most promising drug candidates, helping our industry sponsor Living in Silico Inc., to speed up the drug discovery process using efficient, lab-free methods. Model performance will be measured using F1-score and AUC-ROC.

3. Introduction:

The Problem with Antibiotic Resistance:

In the past few decades,' antibiotics resistance has grown into a big global health issue. Bacteria are evolving faster than the development of new drugs. Because of that, some easy infections due to this bacterium are becoming life-threatening. *Klebsiella pneumoniae* is one of the most dangerous examples. This bacterium is mostly found in hospitals, and due to that, serious infections like pneumonia, bloodstream infections, and urinary tract infections can be found in multiple cases. What makes it especially dangerous is that many strains are now resistant to multiple antibiotics at the same time, a condition known as multidrug resistance (MDR). The World Health Organization has listed MDR K. pneumoniae as one of the highest priority pathogens for new drug development.[1].

Why SSB and PriA Are Good Drug Targets:

To find new antibiotics, we need targets that are essential for bacterial survival but are not found in humans. The interaction between SSB and PriA helicase is therefore a strong candidate target. SSB is a protein that protects single-stranded DNA during replication and repair, while PriA is a helicase that restarts stalled replication forks. Together, they are critical for keeping the bacterial DNA replication process running smoothly. If we disturb this interaction, then the bacteria cannot properly repair or copy their DNA and will not survive.

Importantly, this interaction is highly conserved across many bacterial species, meaning a drug that targets it in K. pneumoniae could potentially work against other dangerous bacteria as well. [2].

Using Computers Instead of the Lab:

Testing thousands of chemical reactions in the lab is expensive, time-consuming and requires multiple resources. Computational drug discovery offers a faster and more affordable alternative. By using machine learning

models, it is easy to research and screen a large number of chemical libraries digitally and identify the most effective drug before any lab work.

Nowadays, Graph Neural Networks (GNNs) have emerged as one of the most powerful tools for this task. Instead of old methods that represent molecules as simple text strings (called SMILES), GNNs treat molecules as graphs with atoms as nodes and chemical bonds as edges. This allows the model to get the actual structure and chemistry of a molecule in a much more meaningful way. [3].

Project Motivation and Sponsor Connection:

This project was developed in partnership with Living In Silico Inc., a company focused on using computational methods to solve real-world problems in infection control and drug discovery.

The goal of our capstone project is to build an accurate model of a GNN-based pipeline that can predict whether a compound will inhibit the SSB-PriA interaction in *K. pneumoniae*. By doing this, we are aiming to provide a practical, scalable tool that reduces the time and cost of early-stage drug discovery and hopefully contributes to the fight against antibiotic-resistant infections.

4. Literature Review

4.1 Project Brief

The capstone project aims at creating a pipeline utilizing deep learning to discover small molecule inhibitors that would bind to the SSB-PriA protein interaction in multidrug-resistant *Klebsiella pneumoniae*. It is undertaken in partnership with Living In Silico Inc., whose mission is to accelerate drug discovery through computational technology.

The fundamental issue is the growing threat of antimicrobial resistance, which is causing conventional antibiotics to fail. The interaction of SSB-PriA is chosen as a new biological target due to its critical participation in the process of replicating and repairing bacterial DNA. Interfering with this interaction will prevent bacterial survival.

The project proposes a solution that uses Graph Neural Networks (GNNs), namely, Graph Convolutional Networks (GCN) and Graph Isomorphism Networks (GIN), on the AID 1272365 dataset (approximately 440,000 compounds). RDKit is used to convert molecules into graph structures, allowing the models to learn their chemical and structural properties.

The contribution that is intended is the creation of a scalable and cost-effective computational screening pipeline that will not rely on costly laboratory experiments.

4.2 Systematic Literature Review.

According to recent research, there has been a significant transition toward the use of AI in drug discovery, as opposed to the classical techniques. [4]. Virtual screening has found extensive use of machine learning models, although most previous methods relied on feature engineering or SMILES-based representations.

Graph Neural Networks have been proposed as a more sophisticated alternative since they encode molecules in the form of graphs, including structural and relational data. With GCN and GIN, it has been established that, when predicting molecular properties and in drug-target interaction tasks, one can obtain good predictive performance.[5].

But the majority of the available studies are, in general, drug discovery studies, as opposed to specifically targeting a protein-protein interaction such as SSB-PriA. [6]. Also, various studies are based on smaller datasets or do not provide a comparative analysis of various GNN architectures. [7]. This clearly shows the need for a targeted, large-scale deep learning approach specifically designed for antimicrobial drug discovery.

4.3 Critical Analysis Table

Reference	Problem	Objectives	Method	Data/Dataset	Results	Limitations	Research Gap	Proposed Improvement
Duvenaud et al. (2015) [8]	Predict molecular activity	Improve prediction accuracy	GCN	Small chemical datasets	Good performance	Small dataset, not scalable	Poor performance on large real-world data	Use a large dataset like AID 1272365
Ramsundar et al. (2015) [9]	Drug screening	Faster prediction	Random Forest / SVM	SMILES data	Moderate accuracy	Cannot capture the molecular structure	Weak representation of molecules	Use graph-based learning (GNN)
Gilmer et al. (2017) [10]	Molecular property prediction	Learn structure	General GNN	Benchmark datasets	Improved results	Not target-specific	No focus on protein interactions	Apply the model to the SSB-PriA interaction
Lim et al. (2019)[11]	Virtual screening	Reduce lab cost	Deep Neural Networks	Medium datasets	Faster screening	No comparison between models	Lack of benchmarking	Compare GCN vs GIN

4.4 Gap Analysis

From the studies reviewed above, a few important limitations can be clearly seen.

First, many studies are based on small or standard benchmark datasets, which do not reflect real-world drug discovery problems. Second, traditional machine learning approaches that rely on SMILES data are not able to properly capture the actual structure of molecules. Third, most GNN-based studies are general and do not focus on specific biological targets. Finally, very limited work has been done on protein-protein interactions, especially the SSB-PriA interaction.

These limitations show that there is a need for a more practical solution that is large-scale, structure-aware, and focused on a specific biological problem.

4.5 Proposed Improvement

This project aims to address the gaps identified above in a practical way.

- A large-scale dataset (AID 1272365) will be used to improve model generalization
- Molecules will be represented as graphs instead of SMILES, allowing better structural understanding
- The model will specifically focus on the SSB-PriA interaction rather than general predictions
- Two models (GCN and GIN) will be compared to find the best approach
- A complete pipeline will be developed, starting from preprocessing (RDKit) to final prediction

This approach is expected to be more realistic and useful for real-world drug discovery.

4.6 Problem Statement

Klebsiella pneumoniae has become a serious health concern due to its resistance to multiple antibiotics. Existing drug discovery methods are slow, expensive, and often not effective. At the same time, current computational approaches either fail to capture molecular structure properly or are not focused on specific biological targets.

Because of this, there is a need for a scalable and accurate deep learning-based solution that can identify inhibitors of the SSB-PriA interaction using molecular graph representations.

4.7 Project Objectives

Based on the problem identified, the objectives of this project are:

- To preprocess molecular data using RDKit and convert it into a graph format
- To implement Graph Neural Network models such as GCN and GIN

- To train and test the models using the AID 1272365 dataset
- To evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and AUC-ROC
- To identify potential inhibitor molecules targeting the SSB-PriA interaction
- To develop a scalable computational tool for drug discovery

4.8 Conclusion

From the literature, AI and deep learning have improved drug discovery, but current approaches still have limitations. Many models are not scalable, do not capture molecular structure properly, or are not focused on specific biological targets.

The main gap identified is the lack of a target-specific, large-scale, graph-based deep learning solution. This project addresses that gap by using GNN models, a large dataset, and a clearly defined biological target.

Overall, this provides a strong foundation for developing an efficient and cost-effective drug discovery approach.

5. Scope of Research

In Scope

- Training models using the AID 1272365 dataset
- Molecular preprocessing using RDKit
- Implementation of GCN and GIN models
- Prediction of SSB-PriA inhibitors
- Model evaluation and comparison
- Fully computational (in silico) approach

Out of Scope

- Laboratory testing or clinical trials
- Full drug development process
- Use of non-graph-based models (beyond comparison)

This keeps the project realistic and manageable within the capstone timeline.

6. Problem Statement

The rise of multidrug-resistant *Klebsiella pneumoniae* has created a strong need for new treatment approaches. Current drug discovery methods are expensive, time-consuming, and often ineffective. In addition, many computational models are not able to properly learn molecular structure.

There is a clear research gap in developing a scalable, graph-based deep learning solution that can accurately predict inhibitors of the SSB-PriA interaction.

This project aims to fill that gap by developing a Graph Neural Network-based pipeline that improves the efficiency and accuracy of drug discovery.

7. Objectives

The main aim of this project is to build a practical and efficient deep learning solution that can help identify molecules capable of blocking the SSB-PriA interaction in multidrug-resistant *Klebsiella pneumoniae*. Since traditional drug discovery methods are slow and expensive, this project focuses on using computational techniques to speed up the process.

To achieve this, the project has the following objectives:

- To use RDKit to preprocess molecular data and convert chemical compounds into graph-based structures
- To design and implement Graph Neural Network models such as GCN and GIN[12]
- To train and test these models using the AID 1272365 dataset, which contains many compounds
- To evaluate how well the models perform using metrics like accuracy, precision, recall, F1-score, and AUC-ROC
- To identify and rank molecules that are most likely to act as inhibitors of the SSB-PriA interaction
- To develop a scalable computational pipeline that can support drug discovery without the need for laboratory testing

8. Limitations

Even though this approach has several advantages, some limitations need to be considered:

- The dataset (AID 1272365) may be imbalanced, which can affect how well the model learns and predicts
- The entire project is based on computational predictions, so there is no experimental validation in a lab
- Limited access to computational resources, especially GPUs, may affect model training and performance
- The accuracy of the results depends on how clean and well-prepared the data is
- Biological systems are very complex, and the model may not fully capture all real-world interactions
- Due to time limitations, it may not be possible to try many different models or perform extensive tuning

9. Methods

This project follows a step-by-step deep learning approach to identify possible inhibitors of the SSB-PriA interaction using Graph Neural Networks. The overall process is explained below.

9.1 Data Collection

For this project, the AID 1272365 dataset is used. It contains screening data for around 440,000 chemical compounds. This dataset is important because it provides labelled information that helps in training machine learning models for drug discovery.

9.2 Data Preprocessing

Before training the models, the data needs to be cleaned and prepared. This is done using RDKit. In this step:

- Invalid or incomplete data is removed

- SMILES strings are converted into graph structures[13]
- Atoms are treated as nodes and bonds as edges[12]
- Useful features are extracted for training

This step is important because better input data usually leads to better model performance.

9.3 Molecular Representation

Each molecule is represented as a graph. This helps the model understand how atoms are connected and how they interact with each other. Compared with traditional methods, this representation captures more meaningful structural information and improves predictive performance.

9.4 Model Development

In this project, two types of Graph Neural Network models are used:

- GCN (Graph Convolutional Network): Helps learn local patterns in molecular structures
- GIN (Graph Isomorphism Network): Better at distinguishing differences between graph structures

These models are designed to classify compounds as potential inhibitors or non-inhibitors of the SSB-PriA interaction.[14]

9.5 Model Training and Validation

The dataset is split into training and testing sets. The models are trained using labelled data, and hyperparameter tuning is performed to improve performance. This also helps ensure that the models work well on new, unseen data.

9.6 Evaluation Strategy

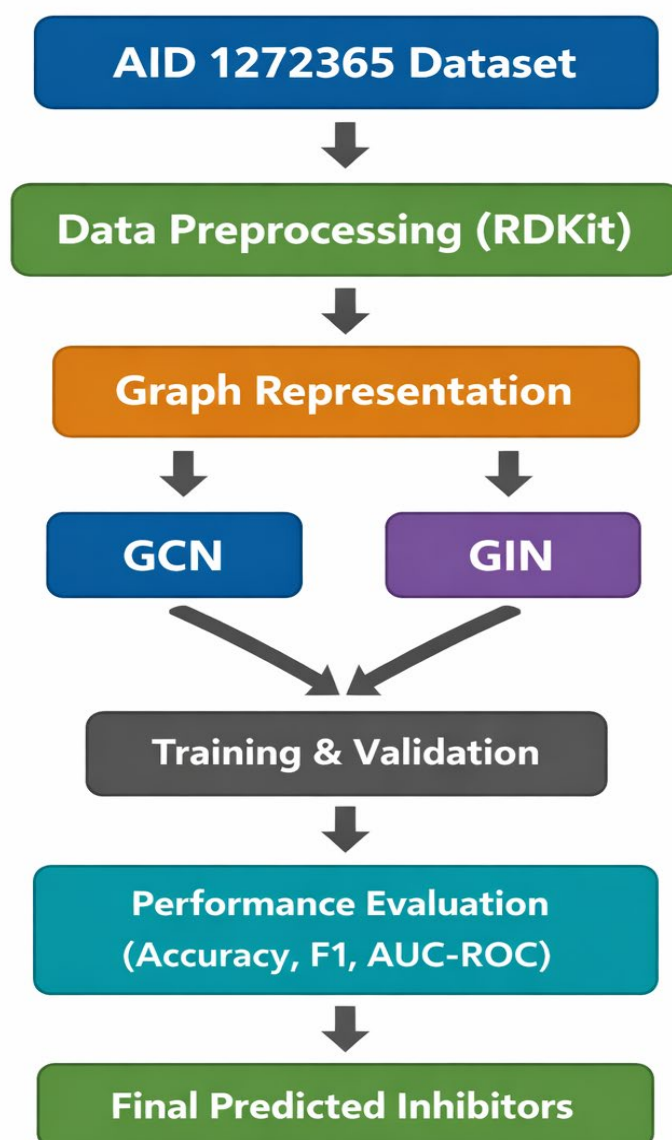
To evaluate the models, the following metrics are used:

- Accuracy
- Precision
- Recall
- F1-score
- AUC-ROC

These metrics help us understand how well the model is performing in identifying potential inhibitors.

9.7 Framework Overview

The overall workflow of the system is shown below:



9.8 Output

The final output of this system is a ranked list of molecules that are predicted to inhibit the SSB-PriA interaction. These results can help the industry sponsor focus on the most promising compounds and save time and cost in the drug discovery process.

10. Database / Datasets

10.1 Primary Dataset - PubChem BioAssay AID 1272365[15]

For this project, we mainly use the PubChem BioAssay dataset called AID 1272365. This dataset aligns closely with our objectives because it focuses on the SSB-PriA interaction in *Klebsiella pneumoniae*.

The data comes from lab experiments where many chemical compounds were tested. Each compound was checked to see how much it could stop the interaction between the two proteins. If a compound showed more than 35% inhibition, we tested it again for confirmation. After that, we labelled it as active or inactive.

This dataset works well for our project for a few simple reasons. First, it is directly related to our problem, so we don't have to modify it much. Second, the labels are already provided, which makes training the model easier. Also, the dataset is large, helping the model learn more effectively. Another advantage is that it is publicly available, so anyone can access it.

Each compound is given in SMILES format. We convert it into a graph using RDKit. In this graph, atoms become nodes and bonds become edges. This makes it easier for the model to understand how the molecule is structured.

10.2 Secondary / Comparative Dataset - ChEMBL

We also looked at the ChEMBL dataset as a secondary source. ChEMBL is a large database that contains information about various bioactive compounds and their behaviour.[16]

The data in ChEMBL comes from research papers and is well-organized. It includes things like binding activity and other biological information. There is also antimicrobial data in it, which is useful for our type of project.

Even though ChEMBL does not specifically focus on SSB-PriA, it is still beneficial. It gives us a broader understanding of antibacterial compounds. In the future, it could help test or compare our model. Since it is open and large, it fits well with the kind of models we are using.

11. Results and Directions

11.1 Expected Outputs

At the end of this project, we expect to get a list of compounds likely to block the SSB-PriA interaction. Both GCN and GIN models will provide a score for each compound. This score indicates how likely the compound is to be active.

We will then sort the compounds based on these scores. The top candidates will be shared with the sponsor, allowing them to focus on the most promising options.

11.2 Evaluation Metrics

To evaluate how well the model is working, we use standard metrics.

First, we examine the confusion matrix:

- True Positive – active compounds correctly predicted
- True Negative – inactive compounds correctly predicted
- False Positive – inactive compounds predicted as active
- False Negative – active compounds predicted as inactive

From this, we calculate:

- Accuracy – overall correctness
- Precision – how many predicted active compounds are actually correct
- Recall – how many real active compounds are found
- F1-score – balance between precision and recall
- AUC-ROC – how well the model separates active from inactive

These help us understand the model's performance clearly.

11.3 Expected Performance

Based on previous studies, we expect both models to perform fairly well. The AUC score should be above 0.80. GIN may perform slightly better than GCN because it handles graph structures more efficiently.

The F1-score may fall between 0.75 and 0.88, depending on how the data is processed.

11.4 Practical Contribution

The main outcome of this project is a functional pipeline that can take raw chemical data and suggest useful drug candidates. This can save significant time compared to testing everything in a lab.

For the sponsor, this means they can focus directly on the best options instead of sifting through thousands of compounds.

11.5 Future Directions

There are several ways to improve this project later:

- The predicted compounds can be tested in a real lab
- The same method can be used for other bacteria
- More datasets can be added to enhance accuracy
- It can be combined with molecular docking
- We can also explore why the model makes specific predictions

12. Conclusion

In this project, we aim to address the real problem of antibiotic resistance. As *Klebsiella pneumoniae* becomes harder to treat, we need faster ways to find new drugs.

Here, we use Graph Neural Networks along with a large dataset to predict useful compounds. By converting molecules into graph form, the model can better understand their structure.

Overall, the idea is straightforward: instead of testing everything in a lab, we first narrow down the best options using a model. This approach can save both time and costs.

The project is practical, the data is accessible, and the method is based on current research, making it ready to move forward.

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